

THORNEY ISLAND, United Kingdom

WMO: 038720

Lat: 50.8147N

Lon: 0.9225W

Elev: 10

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.5	29.6	22.9	17.3	32.7	25.2	19.3	33.4	34.0	49.4	31.2	49.4	6.5	30	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	77.6	66.2	74.3	64.9	71.6	63.4	68.0	75.1	66.2	72.2	64.7	69.8	10.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	93.9	70.8	63.7	88.4	68.8	62.5	84.6	67.5	32.2	75.3	30.8	72.3	29.7	69.9	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.0	26.7	23.6	DB	23.2	82.5	2.5	3.8	21.4	85.2	20.0	87.4	18.6	89.5	16.8	92.2
			WB	23.2	71.3	2.3	3.5	21.5	73.8	20.1	75.8	18.8	77.8	17.1	80.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	42.2	42.3	45.0	49.1	54.5	59.8	63.7	63.5	59.4	54.0	47.8	43.1
	DBStd	9.23	5.68	5.32	4.66	4.91	4.81	3.95	3.86	3.70	3.93	4.99	5.68	6.10
	HDD50	1059	244	217	162	72	13	0	0	0	0	22	106	223
	HDD65	4781	706	635	620	478	328	163	72	72	171	341	516	679
	CDD50	1822	3	2	7	44	151	294	424	418	283	146	40	10
	CDD65	69	0	0	0	0	1	7	31	26	4	0	0	0
	CDH74	312	0	0	0	0	7	44	150	102	9	0	0	0
	CDH80	39	0	0	0	0	0	3	20	16	0	0	0	0
	WSAvg	10.8	12.0	11.2	11.0	10.4	10.7	10.3	10.3	10.0	10.1	11.1	11.2	11.7
	PrecAvg	33.7	3.9	2.5	2.5	2.2	2.0	2.2	2.0	2.4	2.6	3.9	3.7	3.9
(15) Precipitation	PrecMax	45.8	8.2	6.1	6.1	5.2	4.4	6.1	4.3	5.0	5.9	8.3	7.8	7.4
	PrecMin	26.3	0.8	0.1	0.2	0.0	0.4	0.5	0.2	0.2	0.3	1.3	1.0	1.3
	PrecStd	5.1	1.8	1.7	1.1	1.3	1.1	1.6	1.1	1.3	1.5	1.7	2.0	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	54.7	60.3	70.0	74.6	79.2	82.8	82.1	74.9	67.2	60.3	56.3
		MCWB	52.1	50.8	52.4	58.2	62.6	67.0	67.4	68.1	65.2	60.8	58.0	54.1
	2%	DB	52.9	52.6	56.3	63.5	69.9	73.9	78.1	76.4	70.9	64.3	58.5	54.9
		MCWB	51.0	49.8	50.3	55.5	60.5	64.1	66.1	66.3	63.6	60.0	56.1	53.1
	5%	DB	51.6	51.2	53.8	59.4	66.2	70.4	74.5	73.2	68.4	62.7	57.3	53.7
		MCWB	49.9	48.7	48.9	52.7	58.8	62.4	65.2	64.9	61.7	59.2	55.1	52.0
	10%	DB	50.4	49.9	52.0	56.6	62.8	67.7	71.4	70.6	66.4	61.2	56.0	52.3
		MCWB	48.5	47.6	48.4	51.1	56.4	60.9	63.5	63.0	60.6	58.1	53.8	50.5
	0.4%	WB	52.9	52.2	53.7	59.7	64.5	69.3	70.2	71.5	67.4	63.3	58.7	54.8
		MCDB	53.6	53.3	58.7	66.4	71.7	75.9	78.1	77.9	72.1	65.8	59.6	55.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.5	50.6	51.7	56.2	61.5	65.5	67.7	67.9	64.9	61.4	56.9	53.5
		MCDB	52.5	51.8	54.7	62.4	67.7	71.5	75.7	74.6	69.2	63.1	58.1	54.5
	5%	WB	50.3	49.3	50.4	53.8	59.3	63.5	66.0	65.9	63.4	60.1	55.6	52.3
		MCDB	51.5	50.6	52.7	57.9	64.8	69.1	72.9	70.9	66.8	61.9	57.0	53.5
	10%	WB	48.9	48.1	49.1	52.2	57.3	61.7	64.5	64.5	62.0	58.9	54.1	50.7
		MCDB	50.2	49.5	51.2	55.5	61.7	66.6	70.0	69.0	65.2	60.8	55.7	52.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.6	11.6	13.2	13.4	13.5	13.5	13.4	13.5	11.7	10.1	9.2
		MCDBR	8.6	9.7	14.1	17.3	18.3	18.6	19.3	17.4	15.6	11.5	8.8	7.5
		MCWBR	8.3	8.0	9.9	10.9	11.2	11.2	10.5	9.6	9.6	8.3	7.7	7.6
	5% WB	MCDBR	8.5	8.9	11.7	14.6	16.8	16.3	17.1	14.7	13.7	10.0	8.2	7.7
		MCWBR	8.6	8.0	9.0	9.9	10.8	10.7	10.2	9.2	9.5	8.4	8.0	8.1
(40) Clear-Sky Solar Irradiance	taub		0.334	0.350	0.387	0.405	0.408	0.406	0.399	0.393	0.379	0.364	0.337	0.325
	taud		2.369	2.375	2.269	2.247	2.286	2.321	2.346	2.382	2.409	2.423	2.416	2.359
	Ebn at Noon		216	246	255	265	269	270	270	266	257	237	214	199
	Edh at Noon		21	26	35	39	40	39	37	34	30	25	20	18
(44) All-Sky Solar Radiation	RadAvg		263	489	870	1380	1664	1841	1748	1462	1083	613	320	206
	RadStd		20	59	111	131	100	172	139	110	82	50	36	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page